GeonHong Kim

Undergraduate Researcher | Mechanical & Automotive Engineering
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EDUCATION

University of Ulsan Mar. 2022 – Present
Undergraduate Student, Department of Mechanical and Automotive Engineering | GPA: 3.71/4.50

RESEARCH INTERESTS

Wearable Robotics • Soft Robotics • Mechanical Design • Embedded Control Systems • Human-Robot Interaction

RESEARCH EXPERIENCE

Research Intern | InnoCORE Research Internship Program Jun. 2026 – Sep. 2026
University of Ulsan-UNIST, Korea
- Investigated pneumatic air-chamber geometry for attenuating hand-transmitted tool vibration.

Undergraduate Researcher | Human Robotics Lab Dec. 2025 – Present
Department of Mechanical and Automotive Engineering, University of Ulsan, Korea
- Designing and prototyping wearable robotic systems (mechanical components and electronic circuits) and conducting sensor data acquisition.

PUBLICATIONS (DOMESTIC)

Submitted Manuscript Sep. 2026
1. GeonHong Kim, ChanYong Jung, Kyoung-Soub Lee, “Design of a Wearable Robot for Assisting Cable Installation Tasks in the On-Deck Area of Ships”, Transactions of the KSME A.

CONFERENCE PRESENTATION

Poster Presentation May 2026
1. GeonHong Kim, ChanYong Jung, Kyoung-Soub Lee, “Design of a Wearable Robot for Assisting Cable Installation Tasks in the On-Deck Area of Ships”, 2026 Spring Conference of the KSME Ulsan Branch, Ulsan, Republic of Korea.

PATENT

Patent Application in Preparation 2026
1. GeonHong Kim, Kyoung-Soub Lee, “Active Pneumatic Vibration-Damping Glove”

PROJECTS

Active Vibration Damping through Pneumatic Air-Chamber Optimization Jun. 2026 – Sep. 2026
InnoCORE Research Internship, University of Ulsan-UNIST
- Investigated pneumatic air-chamber designs to reduce vibration transmitted from handheld tools to the hand.
- Fabricated Zero-volume Air Chamber (ZAC) and Dielectric Elastomer Actuator (DEA) prototypes to explore alternative actuation mechanisms.

  Role: ZAC/DEA prototyping, Air-Chamber fabrication, IMU-based vibration measurement, and embedded pneumatic actuation.

Soft Robotic Grip-Assist Glove May 2026 – Present
2026 BTS (Brain to Society) Research Program, University of Ulsan-UNIST
- Developing a wearable soft robotic glove to assist gripping during shipyard work.

  Role: Acting as team leader; CAN-based motor communication and control

Active Pneumatic Vibration-Damping Glove Apr. 2026 – Jun. 2026
HRlab, University of Ulsan
- Designed and prototyped a pneumatic glove to reduce hand–tool vibration using air chambers and sensor-based adaptive pneumatic actuation.

  Role: Glove prototyping, IMU-based vibration RMS analysis, pressure-sensor grip detection

1/10-Scale Autonomous Mobility Race Jan. 2026 – Mar. 2026

5th International Collegiate EV Autonomous Driving Competition, International e-Mobility Expo

- Built and integrated a 1/10-scale autonomous vehicle for competition driving tasks.
  Role: Team leader; Chassis design, HW/SW integration, micro-ROS control, and adaptive LiDAR ROI processing.

Cable-Driven Wearable Robot for Cable Installation Dec. 2025 – Present

HRlab, University of Ulsan

- Developing a cable-driven wearable robot to reduce upper-limb physical effort during shipboard cable installation.
  Role: Mechanical design, HW/SW integration, CAN-based motor communication and control.

1/5-Scale Autonomous Driving Platform Jul. 2025 – Sep. 2025

HL FMA 2025-aMAP Innovator Championship, HL Mando

- Developed hardware and embedded control functions for a 1/5-scale autonomous driving platform.
  Role: Hardware design, OpenCR-based control, and LiDAR sensing and control.

AWARDS AND HONORS

Outstanding Paper Award (Poster Presentation) May 2026

2026 Spring Conference of the KSME Ulsan Branch, KSME

Excellence Award (2nd Place), 1/10-Scale Division Mar. 2026

5th International Collegiate EV Autonomous Driving Competition, International e-Mobility Expo

TECHNICAL SKILLS

CAD / Fabrication: Inventor, Fusion 360, 3D printing, silicone fabrication
Embedded / Control: Raspberry Pi, ESP32, OpenCR, CAN motor FOC control
Robotics Software: ROS 2, micro-ROS, Python, C++
Vision: YOLO, OpenCV

CERTIFICATIONS

Craftsman 3D Printer Operator (National Technical Qualification), HRD Korea 2022